Week 7 web page: delivery guide

Instructor notes for Week7_IntervalsAndBayes.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week7_IntervalsAndBayes.html. Open it before class; the widgets need no internet once loaded.
- For live demos, run the matching section of `CODE_Week7_IntervalsAndBayes.R` (same seeds, same numbers). The lab section reuses `nll_pois()`, `haypiles`, `dnll_6`, `lr_cut`, and `lr_int` from the first part, so run the script from the top.
- Zoom the browser to 125-150% for the projector.
- Questions on the page have no answers. Answers are below; *Land:* is where discussion should end up.
- Readings for Canvas: Bolker Ch. 6 through 6.5 (printed pp. 227-267; focus on 6.4.1, pp. 249-254, and 6.5, pp. 260-265) for Monday and the lab; 4.3 (pp. 145-152), 6.2.2 (pp. 237-243), and the Beta distribution, 4.5.2.5 (p. 176), for Wednesday.
- WOD: *Reading the Interval* (in class, Monday), with key. No WOD Wednesday. *Curvature by Hand*, planned as Wednesday's take-home, went out a week later as an optional, uncollected take-home.
- Lab 6: *Reading the Surface* (Act 1, paper) and *An Interval for the Lethal Dose* (Act 2, laptops). Keys: `labs/answer_keys/Lab6_Act1_Key.pdf`, `Lab6_Act2_Key.pdf`.

As taught (Oct. 5-7, 2026)

- **Monday:** the WOD and the big picture, reading questions, curvature, and the likelihood ratio interval (the page's first part, sections 1-4). The planned Bayesian preview, the quadratic approximation, and slices vs. profiles didn't fit. The announcement (added Nov. 4) summarized the 1.92 justification via Wilks' theorem (deviance ~ chi-square when the simpler, nested model is true; df = difference in dimensions; fixing λ removes one); its wording is now in `#curvature` and `#lr-interval`. The project proposal deadline moved to the next Monday.
- **Lab 6:** opened with the quadratic approximation and slices vs. profiles (moved from Monday), then the two Acts. No report on how far groups got. A same-week announcement set the group proposals (due Mon., Oct. 12) and replaced the next lab (Oct. 13) with group meetings; proposal elements and the peer-review plan are now on `FINALPROJ.html#proposal`.
- **Wednesday** (announcement, added Nov. 4): logistic regression and log-odds (Lab 6 had used them first), then Bayesian inference through beta-binomial conjugacy.

The announcement's framing (random parameter vs. fixed; prior x likelihood, normalized) is now in `#bayes-big-picture`; its "compare the beta and binomial formulas; gamma/Poisson too" note is in the conjugacy section. Lab 6 Act 2 was due the next Monday.

- **Wednesday:** about 10 unplanned minutes on logistic regression and the log-odds (Lab 6 had used a logistic dose-response model before lecture introduced it), then the big picture, Bayes' rule on a grid (nests), the Beta-binomial conjugacy algebra, and flat priors on different scales. No discussion period. The germination example, posterior summaries, sticking points, and why Bayes is popular moved to the Week 8 page.
- The page was built after the week (Oct. 29) from the PDF NOTES (now in `lecture_notes/archive/`), the legacy LECTURE4 page (intervals, profiles), and the legacy LECTURE6 page (now in `archive/`). New on the page: the three widgets, the frog brute-force bonus (adapted from LECTURE6), the two Wednesday discussion prompts, and the logistic and grid questions.

Part 1: Uncertainty from the shape of the likelihood (Monday, ~30 min)

WOD (~8 min, before the lecture). Chytrid swabs, 2 of 20 positive, printed NLL curve. About 4 min alone, 2 with a neighbor, 2 debrief.

- 1a: $\hat{p} = 0.10$. 1b: within 2 units, about 0.02 to 0.28 (0.017 to 0.282). 1c: lopsided, 0.08 below and 0.18 above; prevalence can't be negative, and 2 positives already rule out values near 0. 2a: 8 of 80, same $\hat{p}$, curve 4 times steeper, about 0.05 to 0.18.
- Confirm 1a-1b, take one answer to 1c, then: "the rest of today explains where the 2 comes from."
- Page question: "Is the range of plausible values symmetric around the MLE? Why might the curve be steeper on one side?" *Land:* WOD 1c.

Big picture (~3 min)

- Point at FIT. A point estimate is of little use without its uncertainty (Bolker p. 249).
- Same likelihood, two schools: confidence intervals now, posteriors on Wednesday. An interval is the set of values that survive; nothing inside it is proven.

Reading questions (~4 min; not on the page)

- Open the floor. Seeds if it's quiet: "Why are slices 'usually wrong'?" (park for the lab); "Why does more data make the maximum likelihood smaller but the estimate better?" (section 3); "Where does 1.92 come from?" (section 4). Write parked questions in a board corner.

Curvature is information (~6 min)

- Pikas, 6 plots vs. the same counts four times (24 plots). The NLL is a sum, so four times the data is four times steeper.

- Only differences matter: Bolker's coin footnote (1 head in 2 has probability 0.5; 500 in 1,000 has 0.025).

Likelihood ratio interval (~7 min)

- The WOD's 2-unit rule (Edwards), then the LRT version: deviance $\sim \chi^2_1$, 3.84, half is 1.92. Don't derive Wilks; the reference handout has the picture for anyone who asks.
- 6 plots: 1.82 to 4.61 (1.61 above 3, 1.18 below). 24 plots: 2.36 to 3.75, half as wide. Chytrid: 0.017 to 0.278 with 1.92.
- Don't show the Hessian interval or say the chytrid one goes negative; that's the lab's opening.

Discussion (~5 min)

- Q1 ("Is $\lambda = 1.5$ disproven? Is 3 proven?") *Land:* 1.5 is rejected at the 5% level, not disproven (5% of the time a true value lands outside); 3 is the best-supported value, not proven, and 2 and 4 survive too. More data, a different model (overdispersion), or a different confidence level would change the answers.
- Q2 ("Six plots too few to trust the interval?") *Land:* the 1.92 cutoff is a large-sample approximation, so be cautious; check by simulation (simulate many 6-plot surveys from $\lambda = 3$ and count how often the interval covers 3), and check the Poisson assumption itself. A Bayesian interval is another check.

Part 2: Curvature, the Hessian, and two parameters (Lab 6 opening, ~15 min)

Quadratic approximation (~8 min)

- Board: a narrow and a wide parabola with the same vertex. Which has the larger second derivative? Then the Taylor line and SE $= 1/\sqrt{\text{curvature}}$.
- `optim()` Hessian: 2; SE 0.71; Hessian interval 1.61 to 4.39 vs. likelihood ratio 1.82 to 4.61.
- Say plainly: curvature-based SEs are the practical default in multiparameter software; profiles are the check.
- Widget stops, in order (predict before each):
  - 6 plots, mean 3 (default): curvature 2; LR 1.82 to 4.61; Hessian 1.61 to 4.39.
  - 24 plots: curvature 8; LR 2.36 to 3.75; Hessian 2.31 to 3.69.
  - 2 of 20 positive: curvature 222.2; LR 0.017 to 0.278; Hessian -0.031 to 0.231 (shaded, note appears).
  - 8 of 80: LR 0.047 to 0.178; Hessian 0.034 to 0.166.
- "What goes wrong with the Hessian interval for 2 of 20, why can't the likelihood ratio interval do that, and how many frogs until they roughly agree?" *Land:* it runs below 0 because a symmetric parabola ignores the boundary; the LR interval only uses values of p where the likelihood exists, and the true NLL shoots up near 0. At 10% positive:

200 frogs gives 0.064 to 0.147 vs. 0.058 to 0.142; 500 gives 0.076 to 0.128 vs. 0.074 to 0.126. A few hundred.

Slices and profiles (~7 min)

- Tadpoles, Holling II. Point at the tilted valley; slice (purple) vs. profile (green).
- "Which curve would you use for an interval on the attack rate?" *Land:* the profile: the slice pretends h is known, so it's too narrow. Leave it open if the lab hasn't started; Act 1 1a works this out on the beetle surface.
- `solve(hessian)`: SEs 0.071 and 0.0049; correlation 0.82, the tilt in one number.
- **Don't give away:** don't compare $1/\sqrt{H_{aa}}$ with the `solve()` SE (Act 1 2a), and don't show what centering does (Act 1 2b). Don't solve for the LD50 (Act 2).
- **Bonus (why the MLE is normal):** only if asked. Walk the score figure: heights at the truth are normal with variance I; height divided by slope gives variance $1/I$. Point to the reference handout.

Part 3: Logistic regression and Bayes' rule (Wednesday, ~35 min)

Logistic regression and the log-odds (~8 min)

- Odds of 3 is a probability of 0.75. Straight line on the log-odds; e^b multiplies the odds.
- Widget: default ($a = -2$, $b = 0.8$, $x = 2.5$) shows log-odds 0, odds 1, $p = 0.5$. Move x to 3.5: log-odds 0.8, odds 2.23, $p = 0.690$. Then "Slope 0" and "Negative slope."
- "By what factor do the odds change per 100 m? Per 200 m? Does the probability change by a constant factor?" *Land:* odds times 2.23 per 100 m and $2.23^2 = 4.95$ per 200 m; the probability doesn't (0.12 to 0.23 near the edge, 0.88 to 0.94 far out).

Big picture (~3 min)

- Point at FIT again: same model and likelihood, a new answer (a distribution). Prior, likelihood, posterior.

Bayes' rule on a grid (~10 min)

- Nests, three hypotheses: likelihoods 3, 16, 27 (/256), sum 46/256; posteriors 3/46, 16/46, 27/46.
- Grid figure: ask why the bars shrink before saying.
- Widget stops (nest data, 3 of 4):
  - 3 hypotheses: spacing 0.25, tallest 0.587, bar/spacing 2.35, Beta peak 2.11.
  - 11: 0.178 and 2.13. 101: 0.021 and 2.11. 201: 0.010 and 2.11.
  - Click "Probability / spacing": the bars sit on the Beta(4, 2) curve.
- "Which readout keeps shrinking, which settles, and what is the tallest bar with a million hypotheses?" *Land:* the probability per value shrinks like the spacing; bar/spacing

settles at 2.11, the density. A million values gives a tallest bar of about 2.1×10^{-6}. Probabilities for a continuous parameter come from areas.
- Two traps to name: the posterior describes our knowledge of p, not a p that varies; a density of 2.1 isn't a probability.

Conjugacy (~8 min)

- Board: multiply the Beta and binomial kernels, add the exponents. Beta(1, 1) is flat; the nests give Beta(4, 2).
- The constant comes back at the end so the area is 1.
- Flat on p isn't flat on the log-odds: tie back to the logistic section; the figure is on the Week 8 page.

Discussion (~5 min)

- Q1 (201 hypotheses: $P(p = 0.75) \approx 0.01$, "no support"?) *Land:* any single grid value gets a tiny probability that shrinks as the grid gets finer; 0.75 is the posterior mode. Ask about intervals instead: $P(0.65 < p < 0.85) = 0.41$ on any fine grid, and the 95% interval is 0.28 to 0.95.
- Q2 (odds times 2.23 read as "2.2 times as likely") *Land:* odds ratios approximate probability ratios only when success is rare. Near the edge (0.12 to 0.23) it's about 1.9; at 250 m (0.50 to 0.69) it's 1.4; far out the probability can barely rise.

Bonus: brute force with an informative prior. Frog heard on 3 of 10 visits. Flat prior mean 0.333; Beta(15, 5) prior 0.600; ten ponds (30 of 100) with Beta(15, 5), 0.375; Beta(150, 50), 0.600. Conjugacy gives the same numbers. Use it if a student asks what changes when the prior isn't flat.